What is the primary role of a server operating system?
A. To replace network hardware
B. To edit photos only
C. To provide services and resources to client systems
D. To act only as a text editor
ANSWER: C

Which versions are specifically named for installation in the syllabus?
A. Windows Server 2019 or 2022
B. Ubuntu 18.04 only
C. MS-DOS 6.22 only
D. Windows 7 or Windows 8 only
ANSWER: A

What is a server migration?
A. Deleting all user accounts
B. Changing a desktop wallpaper
C. Moving workloads or services from one server environment to another
D. Formatting every client device
ANSWER: C

Why create deployment images?
A. To disable all updates
B. To replace DNS records
C. To increase monitor brightness
D. To standardize and speed up operating system deployment
ANSWER: D

Which activity is part of server lifecycle administration?
A. Only changing icons
B. Only writing documents
C. Installing, upgrading, and maintaining servers
D. Only playing media
ANSWER: C

What should be checked before an OS upgrade?
A. Hardware/software compatibility and backup readiness
B. The desktop theme only
C. The mouse color
D. The number of wallpaper files
ANSWER: A

What is a deployment image?
A. A list of printer queues
B. A DNS query response
C. A screenshot of Task Manager
D. A captured operating system configuration used to deploy systems
ANSWER: D

Why test an image before wide deployment?
A. To avoid naming computers
B. To confirm it installs and works as intended
C. To remove all drivers
D. To make it larger
ANSWER: B

Which practice reduces risk during server migration?
A. Delete the source server first
B. Migrate without testing
C. Ignore dependencies
D. Plan, back up, test, and validate the migration
ANSWER: D

Which is a sensible post-installation step?
A. Erase event logs
B. Remove all admin accounts
C. Disable all security controls
D. Apply updates and verify network and role configuration
ANSWER: D

What is Windows Server Backup used for?
A. Backing up server data and system components
B. Editing registry-free images
C. Creating web pages only
D. Managing Python packages
ANSWER: A

What does a system-state backup help protect?
A. Critical system configuration components
B. Only the monitor settings
C. Only temporary downloads
D. Only browser bookmarks
ANSWER: A

Why is system-state backup important for Active Directory?
A. It prevents all malware
B. It replaces DNS entirely
C. It can support recovery of directory-related system state
D. It changes user passwords automatically
ANSWER: C

What should be done after configuring backups?
A. Disable logging
B. Assume they work without checking
C. Verify that backup jobs complete successfully
D. Delete the backup destination
ANSWER: C

Why store backups separately from the server being protected?
A. To avoid naming folders
B. To reduce the risk of losing both server and backup together
C. To remove restore options
D. To increase CPU usage
ANSWER: B

What is a restore operation?
A. Compressing a file only
B. Recovering backed-up data or system components
C. Changing an IP address
D. Creating a user group
ANSWER: B

Which is a good backup strategy?
A. Back up once and never verify
B. Keep no recovery copy
C. Schedule backups and periodically test restoration
D. Store every copy only on the same failing disk
ANSWER: C

Before restoring a domain controller, what matters?
A. Rename random folders
B. Use a supported recovery procedure and suitable backup
C. Install unrelated games
D. Change the desktop theme
ANSWER: B

What is a backup retention policy?
A. Rules for how long backup copies are kept
B. A user login banner
C. A DNS forwarding rule
D. A rule for screen resolution
ANSWER: A

Why document backup results?
A. To prevent users from logging in forever
B. To replace the backup itself
C. To confirm recovery points and help troubleshoot failures
D. To encrypt all network traffic automatically

ANSWER: C

What is Active Directory Domain Services primarily used for?
A. Centralized identity, authentication, and directory management
B. Disk defragmentation only
C. Local spreadsheet calculations
D. Video editing
ANSWER: A

What is a domain controller?
A. A network cable
B. A client-only workstation
C. A printer driver
D. A server that provides AD DS services for a domain
ANSWER: D

What does an additional domain controller provide?
A. Another domain controller that can support directory services and resilience
B. A replacement for every client
C. A new programming language
D. A backup of the monitor
ANSWER: A

What is a member server?
A. A server that cannot use a network
B. A user's personal account
C. A server joined to a domain but not necessarily a domain controller
D. A domain naming standard
ANSWER: C

What is a key purpose of AD user accounts?
A. Replacing file systems
B. Configuring monitor refresh rate
C. Assigning physical RAM
D. Representing identities that can authenticate and access resources
ANSWER: D

Why use security groups in Active Directory?
A. To manage permissions for sets of users or computers
B. To rename every domain
C. To format disks automatically
D. To store printer ink
ANSWER: A

What does FSMO stand for?
A. File System Management Object
B. Flexible Single Master Operations
C. Flexible Security Mode Operation
D. Fast Server Migration Option
ANSWER: B

Why are FSMO roles used?
A. Certain domain operations are assigned to specific role holders
B. Every client must host all roles
C. They encrypt every file by default
D. They replace all DNS zones
ANSWER: A

What is a common benefit of domain joining a Windows client?
A. Centralized authentication and policy management
B. It disables all updates
C. It removes the need for an IP address
D. It turns the client into a router
ANSWER: A

What should be verified after joining a client to a domain?
A. Connectivity, domain authentication, and policy application
B. Speaker volume only
C. Number of desktop shortcuts

D. Wallpaper only
ANSWER: A

What is the main purpose of DNS?
A. Resolve names to IP addresses and other DNS records
B. Assign passwords
C. Install Windows updates
D. Back up files
ANSWER: A

What is DHCP used for?
A. Encrypt hard drives
B. Automatically provide IP configuration to clients
C. Resolve all file permissions
D. Host Active Directory objects only
ANSWER: B

What does a DHCP scope define?
A. A backup schedule
B. A password length rule
C. A list of Windows services to delete
D. A pool and configuration range for client addresses
ANSWER: D

Why should DHCP exclusions or reservations be planned?
A. To remove routing
B. To stop domain authentication
C. To disable DNS
D. To avoid address conflicts and provide predictable assignments
ANSWER: D

What is IPAM designed to help manage?
A. User photographs
B. IP address space and related infrastructure information
C. Application source code
D. CPU temperature only
ANSWER: B

Which DNS record commonly maps a host name to an IPv4 address?
A. PTR record only
B. MX record
C. TXT record only
D. A record
ANSWER: D

Which DNS record identifies a mail exchanger?
A. A record
B. MX record
C. SOA record only
D. CNAME record only
ANSWER: B

What can happen if DHCP is unavailable and clients have no alternative configuration?
A. All files are deleted
B. The CPU becomes a DNS server
C. Clients may fail to obtain valid network settings
D. The keyboard stops working
ANSWER: C

What is a DHCP lease?
A. A backup file
B. A domain password
C. A time-limited assignment of an IP configuration to a client
D. A Windows service account
ANSWER: C

What should be tested after DNS/DHCP configuration?
A. Name resolution and client address assignment
B. Font installation

C. Printer toner level
D. Wallpaper rotation
ANSWER: A

What is Group Policy used for?
A. Replacing all network protocols
B. Managing only monitor brightness
C. Creating SQL queries
D. Centrally configuring user and computer settings
ANSWER: D

What is a GPO?
A. A backup tape
B. A DNS IP address
C. A type of hard disk
D. A Group Policy Object containing policy settings
ANSWER: D

Where can a GPO be linked in Active Directory?
A. Only to a printer cable
B. To a site, domain, or organizational unit
C. Only to a file extension
D. Only to a browser tab
ANSWER: B

What is a local security policy?
A. Security settings configured on an individual Windows computer
B. A database schema
C. A global Internet standard
D. A DHCP lease
ANSWER: A

Why back up Active Directory?
A. To support recovery from directory data or configuration loss
B. To speed up typing
C. To change all IP addresses
D. To eliminate the need for authentication
ANSWER: A

What is a common reason a GPO may not apply?
A. The keyboard layout is US
B. The monitor is too large
C. The user has too many desktop icons
D. Scope, inheritance, filtering, or connectivity issues
ANSWER: D

What does policy inheritance mean?
A. Applicable settings can flow from higher-level containers
B. The domain loses its OU structure
C. Policies copy files to USB automatically
D. All settings become passwords
ANSWER: A

Why test a new GPO on a limited OU first?
A. To prevent all logins
B. To reduce unintended impact before broader deployment
C. To remove the need for backups
D. To bypass change control
ANSWER: B

What is a restore point in backup planning?
A. A user group
B. A DNS alias
C. A saved state from which data or systems may be recovered
D. A switch port
ANSWER: C

After restoring AD-related data, what should be validated?
A. Only the browser home page

B. Only screen resolution
C. Directory health, authentication, and dependent services
D. Only desktop wallpaper
ANSWER: C

What is IIS?
A. A disk partition format
B. Microsoft's web server platform for Windows
C. A database query language
D. A Linux package manager
ANSWER: B

What is a website binding in IIS used for?
A. Associating a site with protocol, IP/port, and optional host name
B. Creating a disk volume
C. Setting a DHCP lease duration
D. Assigning a user SID
ANSWER: A

Which protocol is commonly used to access a web site hosted on IIS?
A. FTP only in every case
B. ARP only
C. HTTP or HTTPS
D. SMTP only
ANSWER: C

Why configure HTTPS for a web site?
A. To remove the need for updates
B. To increase image resolution
C. To protect traffic using TLS encryption and server authentication
D. To avoid DNS records
ANSWER: C

What is an IIS application pool?
A. A pool of DHCP addresses
B. A list of AD groups
C. An isolation boundary for web applications and worker processes
D. A disk quota
ANSWER: C

What should be checked if an IIS site is unreachable?
A. Only the desktop theme
B. Only the recycle bin
C. Site state, bindings, firewall, DNS, and logs
D. Only the keyboard driver
ANSWER: C

What is a default document in IIS?
A. A DNS root hint
B. A file served when a directory is requested without a filename
C. A Windows license key
D. A backup policy
ANSWER: B

Why use application pool identity appropriately?
A. To bypass authentication
B. To disable all logging
C. To grant every user domain admin rights
D. To limit the privileges used by the web application
ANSWER: D

Which IIS feature helps record web requests for troubleshooting?
A. Disk cleanup
B. Windows Calculator
C. IIS logging
D. Task Scheduler history only
ANSWER: C

What is virtual hosting commonly used for?

A. Replacing Active Directory
B. Creating multiple CPUs
C. Hosting multiple sites on shared server infrastructure
D. Encrypting hard disks
ANSWER: C

How many bits are in an IPv4 address?
A. 64 bits
B. 16 bits
C. 32 bits
D. 128 bits
ANSWER: C

How many bits are in an IPv6 address?
A. 32 bits
B. 128 bits
C. 48 bits
D. 64 bits
ANSWER: B

What is Distributed File System (DFS) used for in Windows environments?
A. Installing graphics drivers
B. Assigning user passwords
C. Running SQL queries
D. Providing a logical namespace and/or replication for shared folders
ANSWER: D

Why are branch-office network solutions important?
A. They remove the need for authentication
B. They eliminate IP addressing
C. They prevent all server maintenance
D. They support connectivity and resource access across locations
ANSWER: D

Which notation is commonly used for IPv4 subnet prefixes?
A. Only host names
B. CIDR notation such as /24
C. Only colon-separated hexadecimal
D. Only MAC notation
ANSWER: B

Which feature is native to IPv6 addressing?
A. A 32-bit address space only
B. A 128-bit address space
C. Mandatory broadcast to all hosts
D. No support for routing
ANSWER: B

What is the purpose of a DFS namespace?
A. Changing disk sectors
B. Presenting shared folders through a logical path
C. Managing user logon hours
D. Assigning DHCP leases
ANSWER: B

What should be considered when designing distributed file access?
A. Network connectivity, permissions, availability, and replication needs
B. Only keyboard language
C. Only wallpaper size
D. Only the number of desktop icons
ANSWER: A

Why test both IPv4 and IPv6 configuration when required?
A. To make all hosts use one address
B. To verify addressing and connectivity for the intended protocols
C. To remove all routes
D. To disable DNS
ANSWER: B

What is a common benefit of DFS replication?
A. Keeping selected folder content synchronized across servers
B. Replacing backups entirely
C. Replicating user passwords to browsers
D. Preventing every network outage
ANSWER: A

What is WDS used for?
A. Managing Linux RPM repositories
B. Network-based deployment of Windows operating systems
C. Monitoring CPU temperature only
D. Creating DNS MX records
ANSWER: B

What is a boot image in WDS?
A. A DHCP reservation
B. A user's profile photo
C. The installed OS image itself in every case
D. An image used to start the deployment environment
ANSWER: D

What is an install image?
A. A network cable diagram
B. A user password database
C. An image containing the operating system to be installed
D. A GPO report only
ANSWER: C

Why is PXE relevant to WDS deployment?
A. It assigns domain roles
B. It allows compatible clients to boot over the network
C. It replaces the boot image
D. It encrypts all files
ANSWER: B

What must be available for network deployment to work?
A. Only a wallpaper file
B. Only a text editor
C. Only a local printer
D. Suitable network services, images, and client connectivity
ANSWER: D

Why test WDS with a pilot client?
A. To avoid all network configuration
B. To validate booting, image selection, and installation before broad rollout
C. To bypass authorization
D. To remove image files
ANSWER: B

What is one benefit of automated OS deployment?
A. No drivers are ever needed
B. Every machine gets a random configuration
C. Consistent and repeatable installation across multiple computers
D. Updates become impossible
ANSWER: C

Which type of client OS deployment is named in the syllabus?
A. Android only
B. Windows 10/11
C. FreeBSD only
D. macOS only
ANSWER: B

What can prevent a client from PXE booting?
A. A missing desktop shortcut
B. A full browser history
C. The screen saver timeout
D. Network boot settings, DHCP/PXE configuration, or connectivity problems
ANSWER: D

After deployment, what should an administrator verify?
A. Only speaker volume
B. Only the lock-screen picture
C. Only browser bookmarks
D. OS installation, drivers, network access, and required configuration
ANSWER: D

What is Hyper-V?
A. A DNS record type
B. Microsoft's virtualization platform
C. A backup compression format
D. A Linux text editor
ANSWER: B

What is a virtual machine?
A. A physical switch
B. A directory permission
C. A user group
D. A software-defined computer running on virtualized resources
ANSWER: D

What is a virtual switch in Hyper-V used for?
A. Configuring Group Policy inheritance
B. Formatting a physical disk
C. Connecting virtual machines to virtual networks
D. Creating user passwords
ANSWER: C

What is a virtual hard disk?
A. A DHCP lease
B. A DNS zone
C. A file-backed or otherwise managed virtual storage device for a VM
D. A user profile
ANSWER: C

Why configure VM memory and CPU settings?
A. To change DNS MX records
B. To allocate appropriate resources to the guest workload
C. To manage printer toner
D. To set domain naming rules
ANSWER: B

What is the purpose of a disk volume?
A. A network protocol
B. A server role for email
C. A logical storage area formatted for use by the operating system
D. A user account
ANSWER: C

What does data deduplication do?
A. Changes file ownership
B. Duplicates every file several times
C. Reduces storage use by identifying and eliminating redundant data blocks
D. Encrypts network traffic
ANSWER: C

What should be considered before enabling deduplication?
A. Only IP address length
B. Only desktop wallpaper
C. Workload support, storage design, and performance requirements
D. Only user names
ANSWER: C

What is a Hyper-V checkpoint commonly used for?
A. Replacing all backups permanently
B. Creating an AD forest
C. Capturing a VM state for a temporary rollback/testing point
D. Assigning public IPs

ANSWER: C

What is a good practice before changing storage layout?
A. Delete all volumes immediately
B. Disable monitoring
C. Verify backups and plan capacity and impact
D. Ignore the filesystem
ANSWER: C

What is FSRM used for?
A. Creating mailboxes
B. Configuring Hyper-V virtual switches
C. Managing DNS root hints
D. Managing and classifying data on file servers, including quotas and file screening
ANSWER: D

What is a disk quota in FSRM?
A. A limit or threshold on storage use for a folder or volume
B. A web binding
C. A DHCP address pool
D. A password policy
ANSWER: A

What is file screening?
A. Controlling or monitoring storage of specified file types
B. Changing a file's extension automatically
C. Scanning network packets only
D. Encrypting all domain accounts
ANSWER: A

Why use storage reports?
A. To replace backups
B. To change IPv6 addresses
C. To create FSMO roles
D. To understand file usage and storage trends
ANSWER: D

What is a hard quota?
A. A DNS record
B. A DHCP lease
C. A warning that never blocks writes
D. A quota that prevents storage use beyond the configured limit
ANSWER: D

What is a soft quota typically used for?
A. Monitoring usage and generating notifications without necessarily blocking it
B. Blocking every file write automatically
C. Removing the file server role
D. Changing user passwords
ANSWER: A

Why define file groups for screening?
A. To configure IPAM scopes
B. To set VM CPU count
C. To create AD domain controllers
D. To group file types for policy application
ANSWER: D

What should be considered when applying file-screening rules?
A. Only the screen resolution
B. Only the DNS suffix
C. Business requirements and legitimate file usage
D. Only the desktop theme
ANSWER: C

How can administrators respond to quota thresholds?
A. Disable the file server
B. Ignore every warning
C. Delete all user accounts

D. Configure notifications or reporting as appropriate
ANSWER: D

What should be tested after FSRM configuration?
A. Quota behavior, file screening, and report/notification delivery
B. Only monitor refresh rate
C. Only the browser home page
D. Only keyboard layout
ANSWER: A

What is Network Policy Server (NPS)?
A. A web browser
B. Microsoft's RADIUS server and network access policy component
C. A disk partition tool
D. A file compression utility
ANSWER: B

What does RADIUS commonly provide?
A. File deduplication
B. Centralized authentication, authorization, and accounting for network access
C. Local printer spooling
D. Virtual disk resizing
ANSWER: B

What is a network policy in NPS used for?
A. Defining conditions and access constraints for connection requests
B. Creating Hyper-V checkpoints
C. Changing server wallpaper
D. Installing a Linux kernel
ANSWER: A

Why must RADIUS clients be configured in NPS?
A. To set disk quotas
B. To create Windows images
C. So NPS can recognize and process requests from those devices
D. To run Python functions
ANSWER: C

What is the purpose of a connection request policy?
A. Changing user display names
B. Assigning DHCP reservations only
C. Determining how connection requests are processed or forwarded
D. Managing file extensions
ANSWER: C

Why protect the shared secret between RADIUS client and NPS?
A. It creates DNS records
B. It changes IP address length
C. It helps secure communication and authenticate the RADIUS client
D. It replaces certificates in all cases
ANSWER: C

What logs help troubleshoot NPS access failures?
A. Only browser bookmarks
B. Only printer queue names
C. Only wallpaper history
D. NPS event and accounting logs
ANSWER: D

What should be validated after NPS setup?
A. Authentication flow, policy matching, and access outcomes
B. Only screen resolution
C. Only file icons
D. Only the desktop clock style
ANSWER: A

What is authorization in network access control?
A. Determining the monitor's size
B. Determining what access an authenticated identity is allowed

C. Formatting the boot disk
D. Creating a software image
ANSWER: B

Why use least privilege in network access policies?
A. To grant only the access required for the role
B. To allow unrestricted access
C. To disable audit logs
D. To make all users administrators
ANSWER: A

What is Network Load Balancing (NLB) intended to do?
A. Create user profiles
B. Distribute client requests across multiple servers
C. Back up Active Directory automatically
D. Replace all DNS records
ANSWER: B

What is a benefit of load balancing?
A. Improved availability and capacity for supported services
B. No need for monitoring
C. No need for server updates
D. Guaranteed zero failures
ANSWER: A

What is a useful first troubleshooting step?
A. Disable all firewalls
B. Reinstall every server immediately
C. Define the symptom and gather relevant logs and configuration
D. Delete all logs
ANSWER: C

Why keep Windows security updates current?
A. To disable authentication
B. To remove all user permissions
C. To avoid backups
D. To address known vulnerabilities and improve security
ANSWER: D

Why audit privileged Active Directory groups?
A. To reduce and detect excessive administrative access
B. To change monitor resolution
C. To assign DHCP leases
D. To increase wallpaper choices
ANSWER: A

What does workstation hardening aim to do?
A. Reduce attack surface and enforce secure configuration
B. Allow unrestricted admin rights
C. Disable all security controls
D. Install every available application
ANSWER: A

Why secure unnecessary Windows services?
A. It automatically fixes all DNS errors
B. It removes the need for authentication
C. Unneeded services can increase the attack surface
D. Services control only screen brightness
ANSWER: C

Which evidence is useful when troubleshooting a service failure?
A. Only desktop icons
B. Only mouse settings
C. Only browser history
D. Service status, event logs, dependencies, and recent changes
ANSWER: D

What is a cluster or NLB health check intended to help determine?
A. Whether a disk label is colorful

B. Whether a node or service is available to handle requests
C. Whether a user's password is memorable
D. Whether a VM has a wallpaper
ANSWER: B

What should be done after a security configuration change?
A. Delete the audit policy
B. Validate functionality and review relevant logs
C. Disable monitoring
D. Assume success without testing
ANSWER: B

What is Microsoft Exchange Server primarily used for?
A. Enterprise email, calendaring, and related collaboration services
B. Disk deduplication
C. Linux package installation
D. IP address management only
ANSWER: A

What is a mailbox in Exchange?
A. A DHCP scope
B. A repository for a user's email and related mailbox data
C. A Hyper-V virtual switch
D. A file-screening rule
ANSWER: B

Why are DNS records important to Exchange services?
A. They create Windows user groups
B. They help clients and mail systems locate relevant services
C. They define CPU allocation
D. They replace storage backups
ANSWER: B

What is a mail flow test intended to verify?
A. That a disk is partitioned
B. That messages can be sent and received as expected
C. That a GPO is linked
D. That a VM has enough RAM only
ANSWER: B

Why plan certificates for Exchange client access?
A. To support trusted TLS connections to services
B. To replace all user accounts
C. To resize virtual disks
D. To create DHCP reservations
ANSWER: A

What should be checked before installing Exchange?
A. Only printer paper
B. Only the desktop theme
C. Only the user's browser bookmarks
D. Supported prerequisites, directory readiness, resources, and deployment requirements
ANSWER: D

What is a mail transport service responsible for?
A. Processing and routing messages through the mail system
B. Creating disk quotas
C. Managing Windows deployment images
D. Running GRUB
ANSWER: A

Why monitor Exchange health after deployment?
A. To change the IP protocol
B. To detect service, mail flow, capacity, and connectivity issues
C. To prevent every possible outage
D. To remove the need for patching
ANSWER: B

What is a mailbox database?

A. A Group Policy Object
B. A DNS zone file only
C. A database that stores mailbox data
D. A virtual switch
ANSWER: C

What is a good security practice for email infrastructure?
A. Expose admin interfaces without controls
B. Disable all auditing
C. Apply updates, restrict admin access, and monitor logs
D. Use shared administrator passwords
ANSWER: C

What is Windows PowerShell?
A. A relational database
B. A command-line shell and scripting language for administration
C. A web browser
D. A disk format
ANSWER: B

What is a PowerShell cmdlet?
A. A hardware device
B. A DNS zone
C. A file permission bit
D. A built-in command following a verb-noun naming convention
ANSWER: D

Which cmdlet commonly lists running processes?
A. Get-Process
B. List-RegistryText
C. Get-DiskWallpaper
D. Show-CPUOnly
ANSWER: A

What is the purpose of try/catch in PowerShell?
A. Changing the shell language
B. Handling terminating errors
C. Creating a DHCP scope
D. Installing a VM automatically in all cases
ANSWER: B

Why use functions in scripts?
A. To replace all cmdlets
B. To remove all error handling
C. To make scripts impossible to test
D. To package reusable logic into named units
ANSWER: D

What is a background job used for?
A. Changing a file extension
B. Running a task asynchronously while the session can continue
C. Creating a user password only
D. Formatting an image
ANSWER: B

What does PowerShell remoting enable?
A. Disabling all authentication
B. Accessing a computer without network connectivity ever
C. Running commands on remote computers
D. Replacing DNS
ANSWER: C

Why should remote administration be secured?
A. It changes the desktop color
B. It reduces the number of commands
C. It removes the need for patching
D. Remote access can expose powerful administrative capabilities
ANSWER: D

What is a useful debugging practice?
A. Disable all logging
B. Delete the script after each failure
C. Inspect errors, variables, and execution flow
D. Ignore exceptions
ANSWER: C

Why reuse and document scripts?
A. To avoid version control
B. To hide what the script does
C. To prevent testing
D. To improve consistency, maintainability, and repeatability
ANSWER: D

Which command displays the current working directory?
A. cat
B. whoami
C. ls
D. pwd
ANSWER: D

Which command lists directory entries?
A. date
B. cd
C. touch
D. ls
ANSWER: D

Which command copies files?
A. mv
B. cp
C. pwd
D. rm
ANSWER: B

Which command moves or renames a file?
A. man
B. grep
C. cat
D. mv
ANSWER: D

Which command changes file permissions?
A. chgrp
B. chmod
C. chown
D. passwd
ANSWER: B

Which command changes file ownership?
A. chown
B. grep
C. find
D. chmod
ANSWER: A

What does an ACL provide beyond basic owner/group/other permissions?
A. DNS name resolution
B. More fine-grained access rules for users and groups
C. Automatic disk formatting
D. Kernel compilation
ANSWER: B

Which command searches text using patterns?
A. cal
B. sort
C. grep
D. touch
ANSWER: C

Which protocol is commonly used for secure remote shell access?
A. Telnet
B. SSH
C. Finger
D. FTP without encryption
ANSWER: B

Why should a disk be partitioned/formatted carefully?
A. They always preserve every file
B. They cannot affect data
C. These operations can destroy existing data
D. They only change the prompt
ANSWER: C

What is Anaconda in many Linux distributions?
A. A Windows domain controller
B. A DNS service
C. A file permission
D. A Linux installation program
ANSWER: D

What is an unattended installation?
A. An installation that uses predefined answers with limited manual input
B. A kernel panic
C. A manual installation requiring every answer
D. A system with no storage
ANSWER: A

What is GRUB used for?
A. Managing user passwords only
B. Searching text files
C. Configuring printer queues
D. Selecting and loading an operating system kernel at boot
ANSWER: D

What is initramfs?
A. A package repository name
B. A network switch
C. A temporary initial filesystem used during early boot
D. A user account
ANSWER: C

What does the boot process do?
A. Only copies user files
B. Only opens a terminal editor
C. Only formats disks
D. Initializes system components and starts the operating system
ANSWER: D

What is the purpose of systemd targets in modern Linux systems?
A. Replacing file permissions
B. Creating RPM packages automatically
C. Encrypting every file
D. Grouping units to represent desired system states
ANSWER: D

Which package format is associated with RPM-based systems?
A. DMG only
B. RPM
C. DEB only
D. MSI only
ANSWER: B

Which package format is associated with Debian-based systems?
A. VHDX only
B. RPM only
C. DEB
D. ISO only

ANSWER: C

Why use a package manager?
A. To edit network cables
B. To change monitor resolution
C. To install, update, remove, and track software packages
D. To assign physical RAM
ANSWER: C

What should be considered before changing bootloader settings?
A. It never needs testing
B. A mistake can prevent the system from booting normally
C. It affects only the desktop wallpaper
D. It always deletes user data
ANSWER: B

What is Kickstart commonly used for?
A. Managing Windows GPOs
B. Creating DNS MX records
C. Automating Linux installations with a configuration file
D. Editing image pixels
ANSWER: C

Why use a Kickstart file?
A. To make repeated installations more consistent
B. To remove package management
C. To replace the kernel
D. To disable all user accounts
ANSWER: A

Which task can be defined in an automated installation configuration?
A. Package selection and installation options
B. Browser bookmarks only
C. Keyboard brand only
D. Monitor brightness only
ANSWER: A

Why should user accounts be configured with least privilege?
A. To disable authentication
B. To give every user root access
C. To avoid file ownership
D. To limit unnecessary access and reduce risk
ANSWER: D

Which command is commonly used to add a local Linux user?
A. useradd
B. userlist
C. mkaccountwin
D. newloginonly
ANSWER: A

Which command is commonly used to change a user's password?
A. tar
B. chmod
C. grep
D. passwd
ANSWER: D

What is a package repository?
A. A boot sector
B. A shell loop
C. A directory permission
D. A source from which package metadata and packages are obtained
ANSWER: D

Why verify package installation results?
A. To ensure required software was installed successfully
B. To disable logs
C. To create a network route

D. To change the filesystem root
ANSWER: A

What is a graceful shutdown intended to do?
A. Delete every package
B. Erase the bootloader
C. Stop services and processes in an orderly way
D. Immediately corrupt open files
ANSWER: C

What should be checked after automated deployment?
A. Only the screen saver
B. Only the hostname font
C. System boot, packages, users, networking, and intended configuration
D. Only the terminal color
ANSWER: C

How many bits are in an IPv4 address?
A. 128
B. 32
C. 16
D. 64
ANSWER: B

How many bits are in an IPv6 address?
A. 128
B. 48
C. 64
D. 32
ANSWER: A

What is OpenSSH used for?
A. Secure remote login and other encrypted network operations
B. Disk partitioning
C. File compression only
D. Managing RPM signatures only
ANSWER: A

What is VNC commonly used for?
A. Package dependency resolution
B. DNS zone signing only
C. Creating local users
D. Remote graphical desktop access
ANSWER: D

Which service typically listens on TCP port 22 for SSH?
A. httpd
B. sshd
C. cups
D. named
ANSWER: B

Why use key-based SSH authentication where appropriate?
A. It disables encryption
B. It removes host verification
C. It grants root access automatically
D. It can provide strong authentication without sending a reusable password each login
ANSWER: D

What is network authentication?
A. Formatting a disk
B. Sorting IP addresses
C. Creating a file archive
D. Verifying the identity of a user or device accessing a network
ANSWER: D

What should be restricted in SSH configuration?
A. All updates
B. All valid user accounts

C. All logging
D. Unnecessary remote access and excessive privileges
ANSWER: D

Which command can show network interfaces and addresses on modern Linux?
A. tar -xf
B. ip address
C. grep -n
D. chmod -R
ANSWER: B

What should be checked if SSH login fails?
A. Only terminal font
B. Only the desktop wallpaper
C. Network reachability, service status, authentication settings, and logs
D. Only file timestamps
ANSWER: C

Which file stores local user account information on many Linux systems?
A. /etc/issue
B. /etc/passwd
C. /etc/fstab
D. /etc/hosts
ANSWER: B

Which file stores password hashes on many Linux systems?
A. /etc/shadow
B. /etc/profile
C. /etc/hostname
D. /etc/group
ANSWER: A

Which command commonly creates a user?
A. usermod
B. userdel
C. groups
D. useradd
ANSWER: D

Which command modifies an existing user account?
A. usermod
B. who
C. useradd
D. pwd
ANSWER: A

Which command deletes a user account?
A. userdel
B. groups
C. useradd
D. id
ANSWER: A

Which command creates a group?
A. chown
B. chmod
C. groupadd
D. groupdel
ANSWER: C

What does the primary group determine?
A. The user's IP address
B. The DNS server
C. The default group association used for a user's files/processes
D. The kernel version
ANSWER: C

What does the id command display?
A. Current time only

B. User and group identity information
C. Disk partition layout only
D. File contents only
ANSWER: B

Why avoid sharing administrator accounts?
A. It removes the need for logs
B. It weakens accountability and auditing
C. It makes permissions more precise automatically
D. It improves traceability
ANSWER: B

What should be reviewed when granting group membership?
A. Only the computer's screen size
B. Only the user's wallpaper
C. Whether the user needs the group's access rights
D. Only the shell prompt color
ANSWER: C

What is fdisk commonly used for?
A. Managing mailboxes
B. Configuring DNS
C. Managing disk partition tables
D. Editing shell scripts
ANSWER: C

What is gdisk commonly associated with?
A. Changing file permissions
B. Starting Apache
C. Creating user groups
D. Managing GPT partition tables
ANSWER: D

What does LVM stand for?
A. Logical Volume Manager
B. Logical Variable Mode
C. Linux Virtual Machine only
D. Local Volume Mount
ANSWER: A

What is a physical volume in LVM?
A. A user account
B. A shell variable
C. Storage device or partition initialized for LVM use
D. A DNS record
ANSWER: C

What is a volume group in LVM?
A. A collection of processes
B. A pool of physical volumes from which logical volumes are allocated
C. A set of network routes
D. A mail server group
ANSWER: B

What is a logical volume?
A. A virtual block device allocated from a volume group
B. A package repository
C. A file permission
D. A kernel module name
ANSWER: A

Why check backups before partitioning?
A. Partitioning always preserves data
B. Backups prevent every hardware fault
C. It changes only file names
D. Partition changes can cause data loss
ANSWER: D

What does mounting a filesystem do?

A. Changes the hostname
B. Makes a filesystem accessible at a directory path
C. Formats the CPU
D. Creates an SSH key
ANSWER: B

Which file commonly lists persistent filesystem mount configuration?
A. /etc/group
B. /etc/passwd
C. /etc/shadow
D. /etc/fstab
ANSWER: D

What should be verified after extending a logical volume?
A. Only DNS MX records
B. The logical volume and filesystem sizes and health
C. Only user shell history
D. Only the desktop theme
ANSWER: B

What is a network interface?
A. A system interface used to connect to a network
B. A disk filesystem
C. A package file
D. A user group
ANSWER: A

Which command can display routing information on modern Linux?
A. tar
B. passwd
C. wc
D. ip route
ANSWER: D

What is a default gateway used for?
A. Forwarding traffic toward destinations outside the local network
B. Managing print jobs
C. Formatting a disk
D. Changing file ownership
ANSWER: A

What is a print spooler used for?
A. Resolving host names
B. Assigning IP addresses
C. Backing up the kernel
D. Queueing and managing print jobs
ANSWER: D

Which Linux service is commonly associated with printing?
A. GRUB
B. NIS only
C. CUPS
D. LVM
ANSWER: C

Why configure a static IP address for some servers?
A. To disable DNS
B. To eliminate all routing
C. To provide predictable network addressing
D. To remove network interfaces
ANSWER: C

What is a subnet mask or prefix used to define?
A. A package version
B. A file owner
C. Which part of an IP address identifies the network
D. A process ID
ANSWER: C

Which utility can test basic IP reachability using ICMP echo requests?
A. chmod
B. touch
C. tar
D. ping
ANSWER: D

What should be checked when network connectivity fails?
A. Only terminal color
B. Only file extension
C. Only user comment
D. Interface status, address, route, DNS, and firewall rules
ANSWER: D

Why manage access to shared print services?
A. To change the filesystem type
B. To assign CPU cores
C. To control who can submit or administer print jobs
D. To configure GRUB
ANSWER: C

Which command is commonly used to manage systemd services?
A. fdisk
B. grep
C. chmod
D. systemctl
ANSWER: D

Which command shows service status in systemd?
A. systemctl status service_name
B. show-dns-service
C. service_name --chmod
D. status all files
ANSWER: A

What is a system configuration file?
A. A monitor profile only
B. A file containing settings that control system or service behavior
C. A CPU register
D. A physical network cable
ANSWER: B

Why back up a configuration file before editing it?
A. To prevent all syntax errors
B. To make the file executable automatically
C. To allow recovery from an incorrect change
D. To remove service dependencies
ANSWER: C

What does NIS traditionally provide?
A. Web hosting only
B. Disk deduplication
C. Network information services for distributing selected identity/configuration data
D. Windows imaging
ANSWER: C

Why is NIS security planning important?
A. NIS encrypts everything automatically
B. Traditional NIS environments can expose sensitive identity information if poorly secured
C. It has no network exposure
D. It replaces all authentication controls
ANSWER: B

Which command can show recent system logs on a systemd host?
A. lsblk only
B. unzip
C. touch
D. journalctl
ANSWER: D

What should be checked if a service fails to start?
A. Only keyboard layout
B. Only wallpaper settings
C. Status output, logs, configuration syntax, and dependencies
D. Only shell history
ANSWER: C

What is a service dependency?
A. A file's color
B. Another unit or resource a service requires to operate
C. A user's preferred editor
D. A DNS suffix only
ANSWER: B

Why apply configuration changes carefully on production servers?
A. It removes the need for testing
B. Configuration never affects service availability
C. A bad setting can interrupt dependent services
D. It always improves security
ANSWER: C

What is a security patch?
A. A network cable
B. A disk partition
C. An update that fixes a security flaw
D. A user group
ANSWER: C

Why apply patches regularly?
A. To prevent all backups
B. To reduce exposure to known vulnerabilities and defects
C. To disable all logs
D. To remove package signatures
ANSWER: B

What is system management?
A. Only editing text colors
B. Only creating directories
C. Maintaining system configuration, services, users, resources, and health
D. Only changing desktop wallpaper
ANSWER: C

What does an X server provide in a traditional X Window System environment?
A. DNS resolution
B. Package repository signing
C. Disk quotas
D. Graphical display services for client applications
ANSWER: D

Why review patch release notes?
A. To remove user groups
B. To understand fixes, dependencies, and potential compatibility impacts
C. To change IP address length
D. To avoid backups
ANSWER: B

What should be done before major patching?
A. Remove the bootloader
B. Assess impact and ensure a recovery plan or backup
C. Disable monitoring permanently
D. Delete logs
ANSWER: B

Which command family is used to install updates on Debian-based systems?
A. fdisk
B. grep
C. chmod
D. apt

ANSWER: D

Which command family is used on many RPM-based distributions?
A. dnf or yum
B. apt-get only
C. CUPS only
D. dpkg only
ANSWER: A

What is a rollback plan?
A. A user group
B. A planned way to return to a working state if a change fails
C. A DNS record
D. A filesystem permission
ANSWER: B

What should be verified after applying patches?
A. System and service health, versions, and relevant logs
B. Only the desktop theme
C. Only file names
D. Only browser history
ANSWER: A

What is DNS used for?
A. Managing print jobs
B. Changing file permissions
C. Resolving names and storing distributed naming records
D. Creating user groups
ANSWER: C

Which DNS record maps a name to an IPv4 address?
A. A
B. TXT only
C. MX
D. PTR only
ANSWER: A

Which DNS record maps a name to an IPv6 address?
A. MX
B. A
C. NS only
D. AAAA
ANSWER: D

Which DNS record identifies an authoritative name server for a zone?
A. NS
B. A only
C. TXT only
D. MX
ANSWER: A

What is a DNS zone?
A. A shell function
B. An administratively managed portion of the DNS namespace
C. A disk partition
D. A user group
ANSWER: B

What is a reverse DNS lookup?
A. Converting a file to a directory
B. Formatting a disk
C. Changing a password
D. Resolving an IP address to a name using reverse records
ANSWER: D

Which command can query DNS records interactively on many Linux systems?
A. tar
B. chmod
C. dig

D. wc
ANSWER: C

What should be checked if DNS resolution fails?
A. Only screen resolution
B. Only keyboard layout
C. Resolver settings, server reachability, zone records, and logs
D. Only file owner
ANSWER: C

What is a DNS TTL?
A. A disk timeout only
B. A Linux process ID
C. The time a record may be cached before it should be refreshed
D. A user login time
ANSWER: C

Why validate DNS configuration before restarting a DNS service?
A. DNS ignores all syntax
B. It formats the disk
C. Syntax errors can prevent the service from loading its zone/configuration
D. It changes the CPU type
ANSWER: C

What is NFS commonly used for?
A. Creating DNS MX records
B. Managing Windows GPOs
C. Formatting virtual disks
D. Sharing filesystems over a network
ANSWER: D

What is an NFS export?
A. A package signature
B. A local user password
C. A server directory made available to NFS clients
D. A shell variable
ANSWER: C

Why configure NFS export permissions carefully?
A. To avoid file ownership
B. To disable all network traffic
C. To restrict which clients and users can access shared data
D. To increase monitor brightness
ANSWER: C

What is FTP used for?
A. Managing systemd units
B. Transferring files over a network
C. Resolving DNS names
D. Creating LVM volumes
ANSWER: B

What is a major security concern with plain FTP?
A. It only works offline
B. Credentials and data may be transmitted without encryption
C. It cannot transfer files
D. It always uses end-to-end encryption
ANSWER: B

What is the purpose of /etc/exports on many Linux systems?
A. Storing DNS cache only
B. Defining NFS exports and their access options
C. Configuring GRUB themes
D. Listing local password hashes
ANSWER: B

What should be checked if an NFS mount fails?
A. Only desktop wallpaper
B. Server export, client permissions, network access, and service status

C. Only shell prompt
D. Only CPU fan speed
ANSWER: B

Why limit FTP access to authorized users and directories?
A. To reduce unauthorized data access and misuse
B. To make all files public
C. To disable logging
D. To replace firewalls
ANSWER: A

What does mounting a remote NFS share do?
A. Attaches the exported filesystem into the client's directory tree
B. Creates a new CPU
C. Deletes the export
D. Changes the IP protocol
ANSWER: A

What should be verified after configuring NFS?
A. Only browser bookmarks
B. Only the hostname font
C. Mount access, permissions, and expected read/write behavior
D. Only keyboard type
ANSWER: C

What is Samba commonly used for?
A. File and printer sharing with Windows-compatible SMB/CIFS clients
B. Linux kernel compilation
C. Disk partitioning
D. Bootloader configuration
ANSWER: A

What is SMB used for?
A. CPU scheduling
B. DNS name resolution
C. Network file and printer sharing
D. Package signing
ANSWER: C

What is a DHCP server responsible for?
A. Serving web pages only
B. Managing shell variables
C. Leasing IP configuration to clients
D. Creating user groups
ANSWER: C

What is a DHCP reservation?
A. A DNS zone transfer
B. A predictable address assignment associated with a client identifier
C. A package dependency
D. A filesystem mount
ANSWER: B

What is the purpose of a DNS server?
A. Answering name-resolution queries for its configured zones or cache
B. Formatting disks
C. Managing print queues
D. Compiling scripts
ANSWER: A

Which configuration file is commonly associated with Samba?
A. /etc/shadow
B. /etc/samba/smb.conf
C. /etc/fstab
D. /etc/hosts.allow only
ANSWER: B

Why align Samba permissions with underlying filesystem permissions?
A. Filesystem permissions never apply

B. Only Samba permissions matter in every case
C. It changes DNS records
D. Both layers can affect whether a user can access shared files
ANSWER: D

What should be checked if a DHCP client gets no address?
A. Only Apache document root
B. Only shell aliases
C. Only file timestamps
D. Scope availability, server status, network path, and DHCP configuration
ANSWER: D

Why test DNS and DHCP after configuration?
A. To resize a disk
B. To change file ownership
C. To confirm name resolution and client address assignment work
D. To install GRUB
ANSWER: C

What is a security best practice for file shares?
A. Make every share world-writable
B. Ignore audit logs
C. Disable authentication
D. Grant only required access and review share permissions
ANSWER: D

What is Apache HTTP Server used for?
A. Managing disk partitions
B. Creating AD users
C. Serving web content over HTTP/HTTPS
D. Assigning IP addresses
ANSWER: C

What is a virtual host in Apache?
A. A virtual disk
B. A DHCP lease
C. A configuration that serves a site based on address/port or host name
D. A user group
ANSWER: C

Which directive commonly sets the web content directory in Apache?
A. DiskVolume
B. DNSRoot
C. DocumentRoot
D. UserGroupOnly
ANSWER: C

Why use HTTPS for an Apache website?
A. To avoid updating software
B. To remove all DNS records
C. To encrypt traffic and authenticate the server with TLS
D. To disable logging
ANSWER: C

What is the purpose of Apache access logs?
A. Storing user password hashes
B. Managing LVM metadata
C. Configuring SSH keys
D. Recording requests received by the web server
ANSWER: D

What should be checked if Apache fails to start?
A. Only the desktop wallpaper
B. Only printer queues
C. Configuration syntax, logs, port conflicts, and service status
D. Only the user's home directory name
ANSWER: C

Why avoid running web applications with unnecessary privileges?

A. To disable HTTP
B. To speed up DNS
C. To replace backups
D. To limit the impact if the application is compromised
ANSWER: D

Which command can test Apache configuration syntax on many distributions?
A. fdisk --test
B. apachectl configtest
C. grep apacheboot
D. chmod testapache
ANSWER: B

What is a common reason two sites need virtual hosts?
A. They must use separate DNS root servers
B. They need different keyboards
C. They need to serve different content or names on shared server infrastructure
D. They need different CPU architectures
ANSWER: C

What should be reviewed after deploying Apache changes?
A. Site response, logs, TLS configuration, and access controls
B. Only terminal color
C. Only screen resolution
D. Only file icons
ANSWER: A

What is Postfix commonly used for?
A. Mail transfer and delivery handling
B. DNS reverse lookup only
C. Web content rendering
D. Disk partitioning
ANSWER: A

What is Dovecot commonly used for?
A. Creating user groups
B. Providing IMAP and POP3 mail access
C. Serving Apache virtual hosts
D. Managing GRUB
ANSWER: B

What is IMAP used for?
A. Formatting filesystems
B. Assigning IP addresses
C. Accessing and managing email stored on a mail server
D. Configuring NTP
ANSWER: C

What is POP3 used for?
A. Running shell loops
B. Creating DNS zones
C. Retrieving email from a mail server
D. Managing Linux packages
ANSWER: C

What is SquirrelMail in the syllabus context?
A. A disk management utility
B. A web-based email client
C. A Linux kernel module
D. A DNS server
ANSWER: B

Why are MX records important for mail delivery?
A. They identify mail exchangers for a domain
B. They assign local file permissions
C. They create system users
D. They configure LVM
ANSWER: A

Why configure TLS for mail services where supported?
A. To replace DNS
B. To disable authentication
C. To change message subject lines
D. To protect credentials and mail traffic in transit
ANSWER: D

What should be checked if outgoing mail fails?
A. Only the file extension
B. Only the terminal prompt
C. Mail logs, DNS/MX records, service status, and relay policy
D. Only the screen saver
ANSWER: C

What is an open mail relay risk?
A. It improves authentication by default
B. It prevents all spam automatically
C. It disables SMTP
D. Unauthorized parties may abuse it to send mail
ANSWER: D

What should be validated after mail server configuration?
A. Only keyboard layout
B. Sending, receiving, authentication, and relevant logs
C. Only browser bookmarks
D. Only disk labels
ANSWER: B

What is performance tuning?
A. Adjusting system or service settings to improve workload behavior
B. Changing only wallpaper
C. Deleting all logs
D. Disabling all services
ANSWER: A

Which command can display running processes and resource use interactively?
A. mkdir
B. top
C. passwd
D. touch
ANSWER: B

Why establish a performance baseline?
A. To avoid monitoring forever
B. To compare current behavior with normal system behavior
C. To replace patching
D. To delete historical data
ANSWER: B

What is a threat model?
A. A structured way to identify assets, threats, attack paths, and mitigations
B. A disk partition map
C. A user shell
D. A package file
ANSWER: A

What is the principle of least privilege?
A. Disable all authentication
B. Share administrator passwords
C. Give only the permissions needed to perform a task
D. Give everyone root access
ANSWER: C

Why monitor CPU, memory, disk, and network usage?
A. To replace backups
B. To identify bottlenecks and unusual behavior
C. To change file ownership
D. To configure GRUB
ANSWER: B

What is a root cause in troubleshooting?
A. A random possible fix
B. The first log line only
C. The underlying reason an issue occurred
D. The hostname alone
ANSWER: C

Why apply defense in depth?
A. Multiple protective layers reduce reliance on one control
B. It disables logging
C. It removes patching needs
D. One control must handle every risk
ANSWER: A

What should be done after maintenance?
A. Assume every change succeeded
B. Verify services, functionality, and system health
C. Disable monitoring
D. Delete logs
ANSWER: B

Which log source is useful for investigating a Linux service problem?
A. Only image thumbnails
B. System and service logs
C. Only browser bookmarks
D. Only desktop icons
ANSWER: B

What is NTP used for?
A. Serving web pages
B. Synchronizing system clocks across hosts
C. Managing disk quotas
D. Assigning IP addresses
ANSWER: B

Why is accurate time important for logs?
A. It changes the file format
B. It replaces authentication
C. It helps correlate events across systems
D. It disables DNS
ANSWER: C

What is BIND commonly used for?
A. A text editor
B. A print spooler
C. DNS server software
D. A partition manager
ANSWER: C

Why restrict DNS zone transfers?
A. To change IP address length
B. To prevent unauthorized parties from obtaining zone data
C. To disable caching
D. To make all DNS queries fail
ANSWER: B

What is a service hardening step?
A. Enable every service
B. Turn off all logs
C. Disable unused features and restrict permissions
D. Share root credentials
ANSWER: C

Why monitor authentication and service logs?
A. To remove the need for access control
B. To create more disk partitions
C. To replace updates
D. To detect failures and suspicious activity

ANSWER: D

What is DNS cache poisoning?
A. Introducing false DNS data into a resolver's cache
B. A user account change
C. A type of disk formatting
D. A print queue error
ANSWER: A

Why use reliable time sources and monitor NTP status?
A. It changes user IDs
B. Clock drift can affect logs, authentication, and scheduled tasks
C. It replaces DNS records
D. It improves wallpaper quality
ANSWER: B

What should be reviewed when securing a network service?
A. Only desktop color
B. Exposure, authentication, permissions, updates, and logs
C. Only the service's icon
D. Only file names
ANSWER: B

What should be done after changing BIND configuration?
A. Disable monitoring
B. Validate configuration and confirm service health and resolution
C. Ignore logs
D. Delete all zone files
ANSWER: B

What is LDAP commonly used for?
A. Serving HTTP content only
B. Compressing files
C. Formatting disks
D. Accessing and managing directory information
ANSWER: D

What is a directory entry in LDAP?
A. A Linux process
B. A disk sector
C. A DHCP lease
D. An object identified within a directory information tree
ANSWER: D

What is NIS traditionally used for?
A. Creating Windows images
B. Managing Hyper-V
C. Distributing selected network identity information
D. Serving web pages
ANSWER: C

What is an Apache cluster intended to support?
A. Availability or scalability across multiple web-server nodes
B. Disk formatting
C. User password hashing only
D. Local shell aliases
ANSWER: A

What is a load balancer used for?
A. Compiling kernels
B. Distributing requests across backend servers
C. Creating filesystem permissions
D. Assigning user IDs
ANSWER: B

Why use TLS with LDAP when appropriate?
A. To change directory names
B. To protect directory credentials and data in transit
C. To replace DNS

D. To disable authentication
ANSWER: B

What is the purpose of NTP in a network?
A. Create web virtual hosts
B. Manage user groups
C. Format volumes
D. Keep system clocks synchronized
ANSWER: D

Why monitor backend health behind a load balancer?
A. To delete server nodes
B. Requests should be sent only to suitable available backends
C. To avoid all network traffic
D. To disable web logs
ANSWER: B

What is a bind operation in LDAP commonly used for?
A. Setting system time
B. Creating a Linux partition
C. Starting Apache
D. Authenticating to an LDAP directory
ANSWER: D

What should be considered when using legacy NIS?
A. Security limitations and whether a more secure identity solution is appropriate
B. It replaces every firewall
C. It encrypts all data by default
D. It never exposes identity information
ANSWER: A

What is a virtual machine?
A. A DNS record
B. A file permission
C. A physical router only
D. A software-based computer running on a virtualization platform
ANSWER: D

What is a hypervisor?
A. A text editor
B. A user account
C. Software that creates and manages virtual machines
D. A package repository
ANSWER: C

What does bridged networking commonly allow a VM to do?
A. Appear as a separate device on the connected network
B. Only communicate with its host in all cases
C. Use no network at all
D. Replace the physical switch
ANSWER: A

What is NAT networking commonly used for in a VM?
A. Creating Linux users
B. Formatting a virtual disk
C. Allowing VM traffic to use a host's network connection through translation
D. Managing file ACLs
ANSWER: C

What is a host-only virtual network commonly intended for?
A. Communication between the host and VMs on an isolated virtual network
B. Disabling VM-to-host communication
C. Public routing to every network by default
D. Replacing DNS everywhere
ANSWER: A

Why configure VM resources carefully?
A. To make all VMs identical regardless of workload
B. To avoid patching

C. To balance guest needs with host capacity
D. To disable snapshots
ANSWER: C

What is a VM snapshot/checkpoint useful for?
A. Capturing a point-in-time VM state for testing or rollback
B. A permanent guarantee against data loss
C. A DNS zone
D. A replacement for all backups
ANSWER: A

What should be considered before reverting a snapshot?
A. It never changes VM state
B. It may discard changes made after the snapshot
C. It only changes wallpaper
D. It always repairs hardware
ANSWER: B

Why isolate lab VMs when testing risky configurations?
A. To limit unintended impact on production or external systems
B. To avoid using IP addresses
C. To disable all learning
D. To remove the hypervisor
ANSWER: A

What should be verified after VM network setup?
A. Only the guest wallpaper
B. Only the VM name
C. Only keyboard language
D. Addressing, routes, DNS, connectivity, and intended isolation
ANSWER: D

What is Bash?
A. A Unix/Linux command shell and scripting language
B. A database server
C. A disk partition format
D. A Windows Group Policy
ANSWER: A

Which symbol redirects standard output to a file and overwrites it?
A. |
B. >
C. 2>&1 only
D. <
ANSWER: B

Which operator appends standard output to a file?
A. ;
B. &&
C. <
D. >>
ANSWER: D

What does the pipe symbol | do in a shell command?
A. Creates a user
B. Sends one command's standard output to another command's standard input
C. Changes file ownership
D. Deletes the first command
ANSWER: B

Which keyword begins a conditional in Bash?
A. if
B. def
C. class
D. selectsql
ANSWER: A

Which loop is commonly used to iterate over a list of values in Bash?
A. catch

B. switchonly
C. for
D. try
ANSWER: C

What does $? usually contain immediately after a command?
A. The system IP address
B. The previous command's exit status
C. The current user's password
D. The number of files in /etc
ANSWER: B

Which command can enable useful Bash execution tracing for debugging?
A. set -disk
B. set -x
C. debugfs -x
D. trace-only
ANSWER: B

Which variable syntax expands a shell variable named name?
A. &name
B. $name
C. %name
D. @name
ANSWER: B

Which utility can test text against regular-expression patterns?
A. mount
B. grep
C. passwd
D. fdisk
ANSWER: B

What is a Bash script?
A. A file containing shell commands for execution
B. A disk partition
C. A DNS zone
D. A Windows GPO
ANSWER: A

What is the usual first line of a portable Bash script?
A. /* /bin/bash */
B. <bash>
C. BEGIN BASH
D. #!/bin/bash
ANSWER: D

Why automate repetitive administration tasks?
A. To remove all validation
B. To prevent logging
C. To make tasks unrepeatable
D. To improve consistency and reduce manual effort
ANSWER: D

What does chmod +x script.sh do?
A. Runs the script once
B. Changes its owner to root automatically
C. Adds executable permission to the script for applicable classes
D. Deletes the script
ANSWER: C

Why should a script check command exit statuses?
A. To hide every error
B. To change the kernel version
C. To detect and respond to failures
D. To remove variables
ANSWER: C

What is a security patch intended to fix?

A. A user shell prompt
B. A desktop wallpaper
C. A known vulnerability or security-related defect
D. A disk label
ANSWER: C

Why schedule and document patching?
A. To prevent all restarts forever
B. To disable package signatures
C. To maintain security while tracking changes and outcomes
D. To avoid testing
ANSWER: C

What should an automated script do if a critical step fails?
A. Continue blindly in every case
B. Delete all logs
C. Handle the failure and report or stop safely as designed
D. Ignore exit codes
ANSWER: C

Why test a script in a lab before production use?
A. To avoid understanding its commands
B. To remove backups
C. To verify behavior and reduce unintended impact
D. To disable permissions
ANSWER: C

What should be reviewed after a patch cycle?
A. Only shell history
B. Update results, reboot requirements, service health, and logs
C. Only file icon size
D. Only desktop color
ANSWER: B

Why is system logging important?
A. It records events useful for troubleshooting and security investigation
B. It removes the need for backups
C. It changes IP addresses
D. It replaces access control
ANSWER: A

Which command can display recent systemd journal entries?
A. journalctl
B. mkdir
C. chmod
D. fdisk
ANSWER: A

What should a monitoring script check?
A. Random unrelated files only
B. Only wallpaper color
C. Only keyboard language
D. Defined service, resource, or event conditions relevant to the system
ANSWER: D

Why include timestamps in monitoring output?
A. To format a disk
B. To establish when an event or check occurred
C. To disable alerting
D. To change file permissions
ANSWER: B

What is log rotation used for?
A. Managing log size and retaining historical logs according to policy
B. Changing user IDs
C. Creating network interfaces
D. Replacing service monitoring
ANSWER: A

Why avoid storing sensitive secrets in logs?
A. Logs may be broadly accessible or retained longer than expected
B. Secrets are never sensitive
C. It improves readability
D. Logs are always encrypted by default
ANSWER: A

What is an alert threshold?
A. A Linux group ID
B. A defined condition that triggers a notification
C. A disk partition table
D. A shell comment
ANSWER: B

What should a monitoring script do when a service is down?
A. Silently ignore it
B. Disable all logs
C. Record the failure and notify or take an approved response
D. Delete the service
ANSWER: C

Why monitor disk space?
A. It creates new users
B. It changes DNS records
C. It increases IPv6 address length
D. A full filesystem can disrupt services and writes
ANSWER: D

What is a useful way to validate a monitoring script?
A. Disable the target service permanently
B. Simulate known conditions and confirm expected logs or alerts
C. Run it once without checking output
D. Delete the logs
ANSWER: B

The company website is inaccessible after an update. Which service should be checked first for an Apache-hosted site?
A. Only the printer queue
B. Apache service status and its logs
C. Only the user's shell history
D. Only the keyboard layout
ANSWER: B

Which command can check Apache service status on a systemd system?
A. fdisk -l only
B. chmod apache status
C. grep -R /dev
D. systemctl status httpd or apache2
ANSWER: D

Remote SSH access fails for some users. What should be reviewed?
A. sshd configuration, authentication, network access, and logs
B. Only Apache DocumentRoot
C. Only the system wallpaper
D. Only disk label
ANSWER: A

A shared directory is inaccessible. Which items should be checked?
A. Only VM wallpaper
B. Only DNS MX records
C. Only Apache port number
D. Ownership, mode bits, ACLs, and parent-directory traversal permissions
ANSWER: D

Which command displays ACLs for a file or directory when ACL tools are available?
A. listports
B. getdns
C. showlvm
D. getfacl

ANSWER: D

Which command changes ACL entries on Linux?
A. setkernel
B. setroute
C. setfacl
D. settime
ANSWER: C

A backup script stopped executing after a change. What is a sensible first step?
A. Disable all backups permanently
B. Delete the script immediately
C. Run it safely with debugging and inspect its error output/logs
D. Ignore its exit status
ANSWER: C

Which command can display the exit status of the last shell command?
A. pwd
B. echo $?
C. echo $HOME only
D. whoami
ANSWER: B

Which log source may help identify why a service failed after an update?
A. Only browser bookmarks
B. System journal and relevant service logs
C. Only wallpaper metadata
D. Only shell aliases
ANSWER: B

After troubleshooting, what confirms the recovery is complete?
A. Delete all logs
B. Disable access controls
C. Restart the server and assume success
D. Test website access, SSH authentication, shared-folder access, backup execution, and logs
ANSWER: D